

PYTHON PRACTICAL 6

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CLASS : SYIT	SUBMISSION : 28/07/26

A . Write a program to compute number of characters and words in a string.

CODE :

```
#Write a program to compute number of characters and
#words in a string.
'''Use split() method for word count. It divides the
string into words using whitespaces as the delimiter
and len() counts the number of elements in the
resulting list.len(s) counts the total number of
characters in the string
(including letters,spaces and punctuation)'''
s="Second Year IT students"
word_count=len(s.split())
Char_count=len(s)

#Printing Result
print("The number of words in a string are :"+str(word_count))
print("The number of characters in a string are :", Char_count)
print("S049 KOMAL ")
|
```

OUTPUT :

```
-----
The number of words in a string are :4
The number of characters in a string are : 23
S049 KOMAL
|
```

B. Create a file geometry.py to calculate base areas for shapes square and circle. In another file, write a function pointyShapeVolume(x, y, squareBase) that calculates the volume of a square pyramid if squareBase is True and of a right circular cone if squareBase is False. x is the length of an edge on a square if squareBase is True and the radius of a circle when squareBase is False. y is the height of the object. First use squareBase to distinguish the cases. Use the circleArea and squareArea from the geometry module to calculate the base areas.

CODE :

File 1. Save as geometry.py

```
*geometry.py - C:/Users/info/OneDrive/Desktop/KOMAL PYTHON/geometry.py (3.14.6)*
File Edit Format Run Options Window Help

def circleArea(r):
    return math.pi * r ** 2

def squareArea(x):
    return x ** 2
```

File 2. Save as any name

```
6c.py - C:/Users/info/OneDrive/Desktop/KOMAL PYTHON/6c.py (3.14.6)
File Edit Format Run Options Window Help

import geometry

def pointyShapeVolume(x, y, squarebase):

    if squarebase:
        base_area = geometry.squareArea(x)
    else:
        base_area = geometry.circleArea(x)
    return y * base_area / 3.0

print("Volume (Square Base):", pointyShapeVolume(4, 2.6, True))
print("Volume (Circular Cone):", pointyShapeVolume(4, 2.6, False))
print("S049 KOMAL ")
```

OUTPUT :

```
S049 KOMAL
Volume (Square Base): 13.866666666666667
Volume (Circular Cone): 43.56341812977846
S049 KOMAL
```